Model Optimization and Tuning Phase Template

Date	July 2024
Team ID	740053
Project Title	Estimating the stock keeping units using Machine Learning
Maximum Marks	10 Marks

Model Optimization and Tuning Phase

The Model Optimization and Tuning Phase involves refining neural network models for peak performance. It includes optimized model code, fine-tuning hyperparameters, comparing performance metrics, and justifying the final model selection for enhanced predictive accuracy and efficiency.

Hyperparameter Tuning Documentation (8 Marks):

Model	Tuned Hyperparameters
Linear Regression	#importing the library for grid search from sklearn.model_selection import GridSearchCV The 'lr_param_grid' specifies different values for regularization strength (C), solvers (solver), and penalty types (penalty). GridSearchCV (lr_cv) is employed with 5-fold cross-validation (cv=5), evaluating model performance based on accuracy (scoring="r2 score"). Linear Regression Hyperparameter Tunning from sklearn.model_selection import GridSearchCV param_grid={'fit_intercept':[True,False],'copy_X':[True,False]} grid_search=GridSearchCV(lr,param_grid,cv=5) grid_search.fit(x_train,y_train) GridSearchCV estimator: LinearRegression LinearRegression pred_cv=grid_search.predict(x_test)
Random Forest	The parameter grid (make_regression) for hyperparameter tuning. It specifies different values for the number of trees (n_estimators), splitting criterion (criterion), maximum depth of trees (max_depth), and maximum number of features considered for splitting (max_features). GridSearchCV (rfc_cv) is employed with 3-fold cross-validation (cv=3), evaluating model performance based on accuracy (scoring="r2 score"). Random Forest Hyperparameter Tunning from sklearn.model_selection import GridSearchCV from sklearn.ensemble import RandomForestRegressor from sklearn.metrics import mean_squared_error from sklearn.datasets import make_regression

Final Model Selection Justification (2 Marks):

Final Model	Reasoning

Random Forest	Random Forest model is chosen for its robustness in handling complex datasets and its ability to mitigate overfitting while providing high predictive r2 score.
	
	Random Forest Regressor
	
	``from sklearn.ensemble import RandomForestRegressor
	from sklearn.metrics import mean_squared_error,r2_score```
	
	```model=RandomForestRegressor()```
	
	``model.fit(x_train,y_train)
	pred=model.predict(x_test)```
	
	``print("Mean Squared Error:",mean_squared_error(y_test,pr
	print("R2 Score:",r2_score(y_test,pred))```
	
	``Mean Squared Error: 892.5601685747586
	R2 Score: 0.7279713962082139```
	
	Above all the models Random Forest model have the highest r2 score among all the models.
	
	A higher r2 score is generally considered better as it indicates a more accurate and reliable model.